Compute Dilution in Budgeted Online RGB-D Gaussian Splatting

Anonymous ICCV submission

Paper ID *****

Abstract

Online RGB-D 3D Gaussian Splatting (3DGS) spends a fixed per-frame optimization budget while its map keeps growing. We study ***compute dilution****: as the map grows, established Gaussians receive a shrinking share of the updates. On two TUM scenes (150 frames, eight seeds each) a corrected-pose selector–throttle factorial, a fourfold-budget control and per-age update logs support this mechanism. Coverage throttling alone improves raw-error selection by* $+3.33$ *and* $+1.77$ dB*; a fourfold optimization budget with the original spawn allowance improves it by* $+3.29$ *and* $+3.07$ dB*; and on* `fr2_xyz` *the fraction of mature Gaussians selected per frame rises from* 0.36 *to* 0.82*. Static map caps do not reproduce the effect on* `fr1_xyz`*, and coverage also changes densification thresholds, so map size is not isolated as the sole cause. A throttled policy with temporal-influence selection reaches* 27.64 dB *on* `fr2_xyz`*,* 2.61 dB *above the strongest budgeted baseline and* 0.21 dB *below unconstrained optimization, with a* $3.8\times$ *smaller map. The effect is scene dependent: the same policy trails raw-error selection on 10 of 14 sequences, including all eight Replica scenes. The* 15 ms *limit is modeled optimization time rather than end-to-end latency (about 10 FPS measured), and seeds measure repeatability on fixed trajectories, not generalization across scenes. Auxiliary analyses (counterfactual Gaussian utility, a learned utility model, contextual re-ranking, and a hole-rate policy router) are in the supplement.*

1. Introduction

3D Gaussian Splatting (3DGS) [6] underlies online RGB-D mapping systems such as SplaTAM [5], Point-SLAM [13], MonoGS [10], Photo-SLAM [4] and RTG-SLAM [12]. Deployed on robots and headsets, these systems have only milliseconds of optimization per streaming frame. At the same time, online pipelines add Gaussians in response to rendering error, so the map grows while the per-frame budget stays fixed; in our runs some maps exceed 100k primitives within 150 frames.

We hypothesize **compute dilution**: a growing map spreads a fixed optimization budget over more Gaussians and over newly spawned ones, so established (mature) Gaussians are updated less often and quality stagnates or degrades. We test this with interventions rather than correlations: a corrected-pose selector$\times$throttle factorial, a fourfold optimization-budget control that holds the spawn allowance fixed, hard map caps, and per-frame logs of how the budget is allocated by Gaussian age. On two TUM scenes the interventions agree with the hypothesis. Throttling spawns raises the per-frame selection rate of mature Gaussians on `fr2_xyz` from 0.36 to 0.82 and improves PSNR by $+3.33$ dB, and quadrupling the optimization budget without touching spawning recovers $+3.29$ dB. The evidence is not clean in every respect: coverage throttling also changes densification thresholds, and a hard cap matching the throttled map size does not reproduce the gain on `fr1_xyz`. We therefore claim a compute-allocation mechanism under the tested protocol, not that map size is the unique cause.

Contributions. (1) A measurable formulation of compute dilution, with age-resolved allocation statistics (budget share spent on young Gaussians, selection rate of mature Gaussians) logged per frame. (2) Interventional evidence on two TUM scenes (eight seeds each): a 2×3 selector–throttle factorial, a fourfold-budget control, and map-cap controls, with their limits stated. (3) A simple coverage-throttled policy that reaches 27.64 dB on `fr2_xyz` within 0.21 dB of unconstrained optimization at a $3.8\times$ smaller map, together with a scene-by-scene account of where it fails (10 of 14 sequences, all Replica scenes). (4) Auxiliary negative and descriptive results (counterfactual single-Gaussian utility, a learned utility model, contextual re-ranking, a hole-rate router), reported in the supplement with frozen artifacts.

2. Related Work

Online 3DGS SLAM and dense mapping. GS-SLAM [19] and SplaTAM [5] introduced differentiable 3D-Gaussian SLAM, with SplaTAM using silhouette-guided optimization; Gaussian-SLAM [21] adds a sub-map design and Loop-Splat [23] adds loop closure by registering Gaussian submaps; Point-SLAM [13] uses dynamic neural point clouds; MonoGS [10] and Photo-SLAM [4] push toward real-time

operation with loop closure; RTG-SLAM [12] adds compact opaque/transparent roles with rule-based stable/unstable selection and is our primary external policy baseline (re-implemented under exactly matched budgets, since native-system comparison would confound renderer, tracker, and scheduler effects). Unlike these systems, our object of study is not the full SLAM stack but the compute-allocation layer inside it; we evaluate mapping under ground-truth poses with tracking explicitly decoupled (Sec. 5). **Scalable and efficient Gaussian mapping.** Recent systems address online map scalability through representation and update design. VTGaussian-SLAM [2] ties Gaussians to RGB-D views and avoids keeping every primitive learnable, targeting large scenes and memory use. SEGS-SLAM [18] improves structure and appearance consistency, while DyGS-SLAM [3] focuses on dynamic-scene localization and reconstruction. In parallel, SteepGS [17] derives compact density-control updates for 3DGS reconstruction. A recent online-mapping preprint studies geometry-aware initialization and densification under short optimization schedules [8]. These methods make clear that map representation, density control, and online optimization are coupled. Our study asks how growth rate and update policy should respond to coverage and observable scene statistics under a fixed per-frame budget and Gaussian representation. We do not claim density control or efficient Gaussian mapping as new in general. The per-scene SOTA numbers in Appendix Supp. Tab. S2 are protocol-mismatched context. **Density control for 3DGS.** Taming 3DGS [9] uses guided, constructive densification under resource limits; Mini-Splatting [1] targets a constrained Gaussian count; Pixel-GS [22] reweights the densification gradient by covered pixels; and 3DGS-MCMC [7] replaces heuristic densification and pruning with relocation. These methods address offline reconstruction. Our setting differs in that a per-frame optimization budget is shared between existing and newly spawned Gaussians. We do not benchmark against them, which is a limitation of the present comparison (Sec. 5). **Evaluation protocols in dense SLAM.** Published systems report full-sequence train-view rendering with global refinement at full resolution (Replica 33–42 dB for MonoGS/Point-SLAM/RTG-SLAM; TUM fr2_xyz 16–25 dB for MonoGS/SplaTAM/Photo-SLAM) plus full-trajectory ATE. Our protocol is deliberately narrower — budgeted online mapping on fixed horizons — so cross-paper numbers are contextual range, never superiority claims (Supp. Tab. S2). **Budgeted and utility-driven selection.** Error-only, error×influence, gradient-norm, and temporal-filtered heuristics all trade quality for compute. We benchmark them under identical modeled budgets with paired Holm-corrected tests rather than reporting best-of-sweep numbers. Their ordering differs between the evaluated TUM and Replica scenes (Sec. 4). **Submodular maximization and knapsack allocation.** Budgeted set selection is related to monotone submodular maximization, for which approximation guarantees exist under explicit assumptions and appropriate algorithms [11, 16]. Those assumptions are not established for our measured reconstruction objective, and our fractional-knapsack ranker does not inherit those guarantees. The measured interactions vary by protocol, so we treat pointwise ranks as references and test map-growth control empirically.

3. Setting and Method

Budgeted mapping. Let $G_t = \{g_1, \ldots, g_N\}$ be the active map, $g_i = (\mu_i, \Sigma_i, \alpha_i, c_i)$, with streaming observation (I_t, D_t) and pose T_t. Per frame, a scheduler chooses a subset $A_t \subseteq G_t$ to optimize,

$$\max_{A_t \subseteq G_t} \Delta Q(A_t) \quad \text{s.t.} \quad \sum_{i \in A_t} C_i \leq B_t, \tag{1}$$

where ΔQ is the reconstruction improvement, approximated at runtime by a selector score, and $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$ is a decoupled cost model for K microsteps. The budget B_t is *modeled* optimization time. **Measuring dilution.** We log two age-resolved statistics per frame: the *young budget share*, the fraction of modeled update budget spent on Gaussians younger than ten frames, and the *mature selection rate*, the fraction of Gaussians at least ten frames old that are selected. Dilution predicts a falling mature selection rate as the map grows. **Selection.** Candidates are scored by error × influence mass, optionally with temporal EMA filtering ($\beta = 0.90$), and packed by fractional-knapsack value density s_i/C_i under $B_t = 15\,\text{ms}$ with safety factor 1.10 (effective cap 13.6 ms). Each selected Gaussian receives $K = 5$ microsteps; the cost model is calibrated so modeled and wall-clock optimization time agree within measurement noise on the RTX 4050 backend. Exact Gaussian-to-pixel attribution costs 523.9 ms/frame, so we use a center-sampled projection, $\tilde{m}_i = \alpha_i \det(\Sigma_{2D,i})^{1/4}\, e(\lfloor \mu_{2D,i} \rceil)$, that runs in 0.85 ms (614.5× faster) at Spearman $\rho = 0.9079$ and Pearson $r = 0.9028$ against exact attribution. The scheduler totals 2.54 ms per frame. **Coverage throttling.** Coverage κ_t is the fraction of valid pixels with transmission < 0.5. The per-frame spawn cap ramps from 1.0 to 0.20 as κ_t goes from 0.80 to 0.90. A backlog limit exists but was inactive in the headline runs, because its count was only computed when warmup was enabled; the factorial evaluates it independently. Warmup pre-allocation is disabled and microsteps are fixed at $K = 5$.

4. Experiments

Protocol. TUM RGB-D [15] and Replica [14] sequences at 320 × 240, $B_t = 15\,\text{ms}$, seeds [42..49]; two-sided paired Wilcoxon tests, Holm correction and bootstrap CIs over per-seed trajectory summaries; production `gsplat`

backend; ground-truth poses with tracking decoupled (a mapping benchmark, not SLAM ATE). With eight seeds the smallest attainable two-sided Wilcoxon p is 0.0078, and with fixed trajectories seeds measure repeatability only, so we report effect sizes with intervals and do not treat seed-level standardized effects as evidence of generalization. Artifacts: `results/phase14_projected_area_fix_clean_7policy_8seeds/`. **Holdout protocol.** Because online mapping optimizes on the evaluated trajectory, train-view PSNR overstates generalization. We hold out every 8th frame render-only (never mapped, never selected): the train–holdout gap is $+0.47$ dB mean over 30f×3 seeds for both OURS and `error_only` (`results/phase14_holdout/`). This is a same-trajectory proxy, not independent novel-trajectory generalization.

Main result (Tab. 1, Fig. 2, Fig. 3). OURS reaches 27.64 ± 0.04 dB: $+2.61$ over the strongest budgeted baseline, $+4.48$ over the RTG-SLAM-policy re-implementation, and $+8.58$ over pass-through. These results evaluate the combined policy on this trajectory.

Selector × throttle factorial (Tab. 2). With selector fixed to raw error, coverage adds $+3.33$ $[3.19, 3.50]$ dB on `fr2_xyz` and $+1.77$ $[1.73, 1.81]$ dB on `fr1_xyz`. With temporal×influence fixed, its gains are $+2.68$ $[2.64, 2.72]$ and $+1.42$ $[1.41, 1.44]$ dB. Brackets are paired seed-bootstrap 95% CIs; all four coverage contrasts have two-sided paired Wilcoxon $p = .0078$. Temporal selection itself adds $+0.26$ dB with coverage on `fr2_xyz` and $+0.59$ dB on `fr1_xyz`; the interaction is negative (-0.65 and -0.35 dB), so selector and coverage effects are not additive. The independent backlog option adds $-0.02/+0.02$ dB on `fr2_xyz` and only $+0.07/+0.07$ dB on `fr1_xyz` for error/temporal selection; on the latter scene its lower map size accompanies worse depth L1.

Dilution controls and direct allocation. Holding the spawn allowance at its 15 ms value, giving raw-error selection 60 ms for optimization raises PSNR by $+3.29$ $[3.18, 3.41]$ and $+3.07$ $[3.04, 3.11]$ dB on `fr2_xyz`/`fr1_xyz`, despite maps of 94k/79k. Capping the error-only map at 23k adds $+2.64$ $[2.53, 2.76]$ and $+1.18$ $[1.15, 1.22]$ dB, but doubles depth L1 on `fr1_xyz`. A 39k cap matching that scene's coverage map adds only $+0.04$ $[-0.06, 0.11]$ dB and also harms depth. Thus a static size limit does not reproduce coverage throttling's effect on `fr1_xyz`. For raw-error selection on `fr2_xyz`, coverage reduces the share of modeled update budget spent on Gaussians younger than ten frames from 22.5% to 10.7%, while the per-frame selection rate among mature Gaussians rises from 35.8% to 82.3%. The corresponding mature selection rate on `fr1_xyz` rises from 28.6% to 56.3%. Allocation shifts and the budget intervention support compute dilution under the tested protocol, but coverage also changes densification thresholds and the optimization trajectory. Map size alone is not identified as the unique cause.

Exploratory transfer to two more TUM scenes (150f×3 seeds, same six factorial cells): coverage adds $+0.34/+0.10$ dB to error/temporal selection on `fr1_desk`, but depth L1 worsens from 0.0481/0.0476 to 0.0590/0.0595 m. On `fr3_static`, coverage adds $+0.77/+0.38$ dB and slightly improves depth L1. Adding backlog to temporal selection on `fr1_desk` loses 0.59 dB. These scene-level differences limit any claim that one growth rule is optimal across sensor and motion conditions; the three-seed comparisons are descriptive (full traces in `results/dilution_factorial/`).

Observed failure cases. Two TUM scenes show where the combined policy trails raw-error selection. *Exploration-heavy motion* (`fr1_room`, -1.04 dB) coincides with a smaller final map (92k vs 97k); the available run does not isolate whether throttling or selection causes the gap. *Rapid rotation* (`fr1_rpy`, -1.13 dB mean, finals 24.5 vs 24.1) is consistent with a lag from temporal smoothing, but this attribution has not been tested independently. **Horizon effect (Fig. 1):** at 30 frames OURS ties `error_only` (-0.01 dB on `fr2_xyz`); the gap appears at longer horizons. The new age-group logs in Tab. 2 link this divergence to a smaller update share for mature Gaussians in the growing map. Around frame 65 in seed 42, both error-only variants encounter difficult views; the unthrottled run continues spawning roughly 500–900 splats per frame and its PSNR remains low after frame 70, whereas the coverage run spawns roughly 90–130 and recovers by frame 75. The unthrottled loss rises from about 0.07 at frame 65 to 0.13 at frame 70 and stays high; opacity declines while the map expands. This is sustained divergence following shared hard views, not evidence for a single isolated gradient explosion. The logs cannot rule out floaters without a spatial geometry audit.

Scene dependence (Tab. 3). The combined policy wins the two confirmatory TUM comparisons ($n = 8$): `fr2_xyz` $+3.59$ dB and `fr1_xyz` $+2.38$ dB over raw-error selection, both $p = 0.0078$. It trails raw-error selection on all eight Replica scenes (-1.93 to -4.21 dB; `office0` and `office1` at $n = 8$, the rest exploratory at $n = 3$) and on `fr1_room` and `fr1_rpy`, so across 14 sequences it loses mean PSNR on 10. On Replica `office0`, raw-error selection with a weak throttle ($\kappa = 0.98$) reaches 30.17 dB versus 26.85 dB for the combined policy (supplement). We did not run the selector×throttle factorial on Replica, so we cannot say whether the Replica loss comes from the temporal-influence selector, from the stronger throttle, or from the scene regime; the dilution evidence is limited to the two TUM scenes. A simple hole-rate router that picks the better arm on these two datasets is described in the supplement; its within-scene depth interventions did not reverse the ranking, so we do not use it as evidence for any mechanism.

Policy	Mean PSNR	Final PSNR	SSIM	FPS	Map	Δ vs OURS
no_op	19.06	14.88	0.6920	90.0	23,523	+8.58 ✓
error_only	24.04	20.12	0.8838	9.8	73,383	+3.59 ✓
error_influence	25.03	21.83	0.9083	9.5	105,438	+2.61 ✓
error_influence_temporal	24.94	21.61	0.9079	9.5	106,392	+2.70 ✓
rtg_slam_reimpl	23.16	20.04	0.8669	8.9	65,960	+4.48 ✓
OURS	**27.64**	**28.82**	**0.9231**	**11.1**	**23,765**	—
FULL	*27.85*	*29.34*	*0.9245*	*9.6*	*90,582*	−0.21 ✓

Table 1. **Main benchmark, TUM `fr2_xyz`, 150 frames × 8 seeds.** All six controls differ significantly from OURS after Holm correction ($p_{\text{Holm}} = 0.0469$). FULL is 0.21 dB higher. Headroom recovery is 97.6% relative to no-op and 92.5% relative to the strongest budgeted baseline, at 3.8× smaller map than FULL.

	TUM fr2_xyz			TUM fr1_xyz		
Selector / map control	**PSNR**	**Depth L1**	**Map (k)**	**PSNR**	**Depth L1**	**Map (k)**
Error, no throttle	24.05	.0236	80.3	21.12	.0505	80.6
Error, coverage	27.38	.0148	24.4	22.89	.0443	37.9
Error, coverage + backlog	27.36	.0149	24.4	22.95	.0456	35.9
Temporal×influence, no throttle	24.96	.0165	105.5	22.06	.0484	89.1
Temporal×influence, coverage	27.64	.0130	23.8	23.48	.0425	38.7
Temporal×influence, coverage + backlog	27.66	.0131	23.4	23.55	.0450	32.5
Error, optimization budget ×4	27.34	.0142	93.8	24.19	.0402	79.3
Error, hard map cap 23k	26.69	.0215	23.0	22.30	.1023	23.0
Error, hard map cap 39k	—	—	—	21.15	.1046	39.0

Table 2. **Corrected-pose factorial and dilution controls, 150 frames × 8 seeds per scene.** All factorial cells use the same modeled 15 ms optimization budget. The ×4 control uses 60 ms for optimization while keeping the original 15 ms spawn allowance. Depth L1 is in meters. Hard caps harm depth on `fr1_xyz`; the 39k cap matches the coverage map size there but gives no PSNR gain. Backlog is enabled independently of warmup only in the last throttle level. Per-run traces and paired intervals are in `results/dilution_factorial/`.

Systems. Scheduler overhead is 2.54 ms; optimization-budgeted execution runs at about 10 FPS, so this is *not* end-to-end 15 ms real time. A separate baseline odometer (dense photometric+depth Gauss-Newton, no keyframes or loop closure) reaches ATE 0.9/2.0 cm on `fr2_xyz`/`fr1_xyz` over 100 frames (Supp. Tab. S3); it is a sanity check, not a SLAM claim. AlexNet LPIPS on `fr2_xyz` (150f×3) is 0.211 for OURS versus 0.294 for `error_only`; depth L1 is in Tab. 2, and we report no mesh-based surface accuracy. Published-system numbers use different frames, resolution and views and give range context only (Supp. Tab. S2); short-horizon budget-curve and coverage-knee ablations are in Supp. Figs. S1–S2. **Reproducibility.** Every table maps to a frozen bundle and a command in `paper_repro/`; the headline bundle records per-policy configs, source hashes, artifact hashes, pose protocol and clean-worktree status of each run set. Prior pose-bugged numbers (19.24 dB) are quarantined in `results/final_confirmation/DEPRECATED.md` and never compared. Environment: RTX 4050 Laptop GPU (6 GB), PyTorch 2.11+cu128, `gsplat` [20] 1.5.3, Python 3.12; peak allocation stays under 3.5 GB with the desktop compositor active, which is the intended deployment envelope for budgeted online mapping.

5. Limitations

Scope of the mechanism. The factorial, budget control and allocation logs cover two TUM scenes. Coverage throttling changes both spawn allowance and error thresholds; neither the map caps nor the traces isolate map size as the sole mediator. The 23k cap improves RGB on `fr1_xyz` but doubles depth error, while a 39k cap gives no RGB gain. On Replica the combined policy loses, and we have no factorial there. Additional TUM scenes are exploratory ($n = 3$) and show smaller or depth-degrading effects. **Protocol.** Mapping uses ground-truth poses; horizons are at most 150 frames (about five seconds at 30 FPS); PSNR is train-view with only a same-trajectory holdout proxy. The 15 ms budget is modeled optimization time, not frame latency, and the pipeline runs at about 8–11 FPS on a laptop GPU. We do not compare against offline density-control methods [1, 7, 9, 22] or integrate the policy into a full SLAM system with its own tracker. **Statistics.** Eight seeds establish repeatability on fixed trajectories; the unit of generalization is the scene, and only two scenes are confirmatory. **Open items.** A Replica factorial, longer sequences with a real tracker, comparisons against density-control baselines, and a continuous (map-size-proportional) budget intervention would test whether

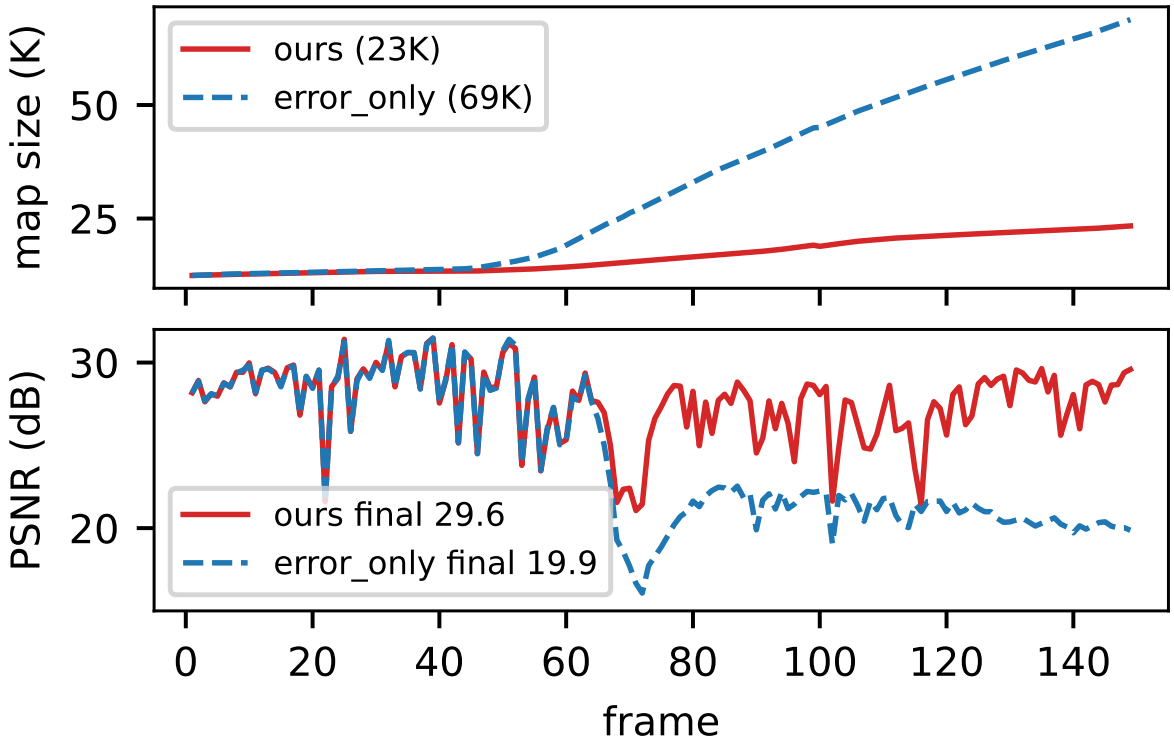


Figure 1. **Map growth and reconstruction quality over time** on TUM `fr2_xyz` and `fr1_xyz` (150 frames, eight seeds). Lines show the seed mean and shaded bands show ±1 standard deviation. `error_only` versus `error + coverage` holds the selector fixed to show the coverage effect; `temporal + coverage` adds the combined policy. Paired sequence-level contrasts are reported in Tab. 2.

Scene	Frames	OURS	`error_only`	Δ	Map (k)	n
fr2_xyz (TUM)	150	27.64	24.04	+3.59 ✓	24 / 73	8
fr1_xyz (TUM)	150	23.47	21.09	+2.38 ✓	39 / 81	8
fr1_desk (TUM)	150	22.55	22.34	+0.21	27 / 56	3
fr1_room (TUM)	100	15.53	16.54	−1.01	92 / 97	3
fr1_rpy (TUM)	100	19.92	21.06	−1.14	45 / 62	3
fr3_static (TUM)	100	25.96	25.88	+0.08	13 / 18	3
office0 (Rep.)	150	26.85	30.17	−3.32 ✓	79 / 116	8
office1 (Rep.)	100	29.25	33.46	−4.21 ✓	42 / 74	8
office2 (Rep.)	99	22.08	24.67	−2.59	70 / 117	3
office3 (Rep.)	99	22.45	24.47	−2.02	77 / 150	3
office4 (Rep.)	99	23.58	25.93	−2.36	81 / 141	3
room0 (Rep.)	99	20.85	22.78	−1.93	82 / 150	3
room1 (Rep.)	100	20.42	24.29	−3.87	47 / 59	3
room2 (Rep.)	99	22.86	25.88	−3.03	78 / 146	3

Table 3. **Per-scene combined-policy breakdown, 14 sequences.** ✓ marks the reported significant comparisons at $n = 8$; other entries are exploratory. The combined policy wins the two confirmatory TUM comparisons but loses on both confirmatory Replica comparisons; remaining scenes have exploratory sample sizes.

dilution generalizes beyond the TUM scenes studied here.

6. Conclusion

Under a fixed modeled optimization budget, interventions on two TUM scenes support *compute dilution*: throttling

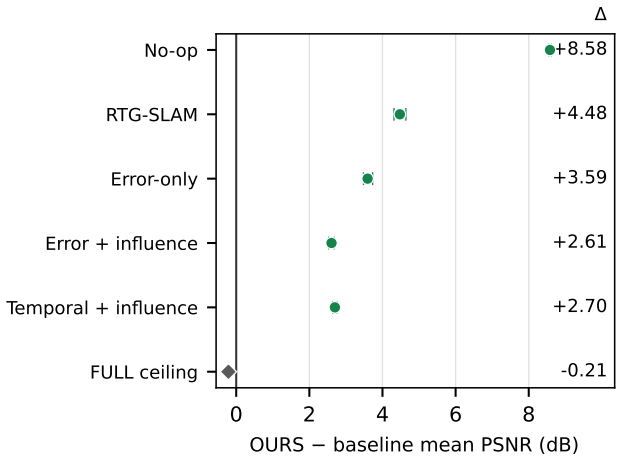


Figure 2. **Paired mean-PSNR differences** on fr2_xyz (150 frames, $n = 8$). Points show OURS − baseline; bars are 95% paired-bootstrap CIs. Positive favors OURS. FULL is 0.21 dB higher; all six contrasts pass Holm correction ($p = 0.0469$).

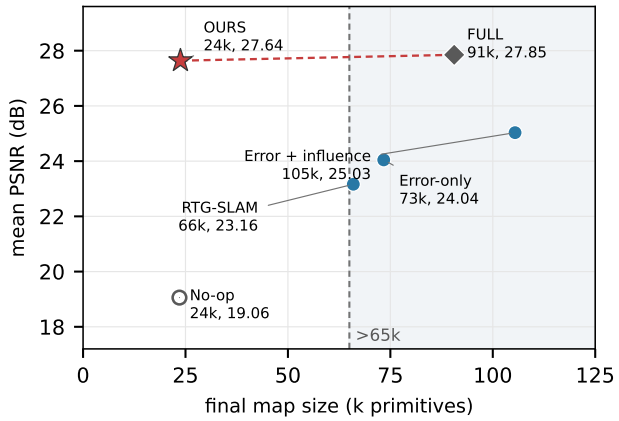


Figure 3. **Quality–map-size trade-off** on fr2_xyz (150 frames, $n = 8$). Only OURS and FULL lie on the frontier; FULL gains 0.21 dB with a 3.8× larger map. The three growing heuristics exceed 65k; no-op stays compact at lower quality.

spawns raises the update share of mature Gaussians, and quadrupling the budget without changing spawning recovers a comparable PSNR gain, while static map caps reveal additional geometry and trajectory effects. A coverage-throttled policy with temporal-influence selection reaches 27.64 dB on `fr2_xyz`, 0.21 dB below unconstrained optimization at a 3.8× smaller map, but it loses on most other sequences, notably Replica. We therefore present dilution as a mechanism to account for, with a scene-dependent remedy, rather than as a universally optimal growth rule.

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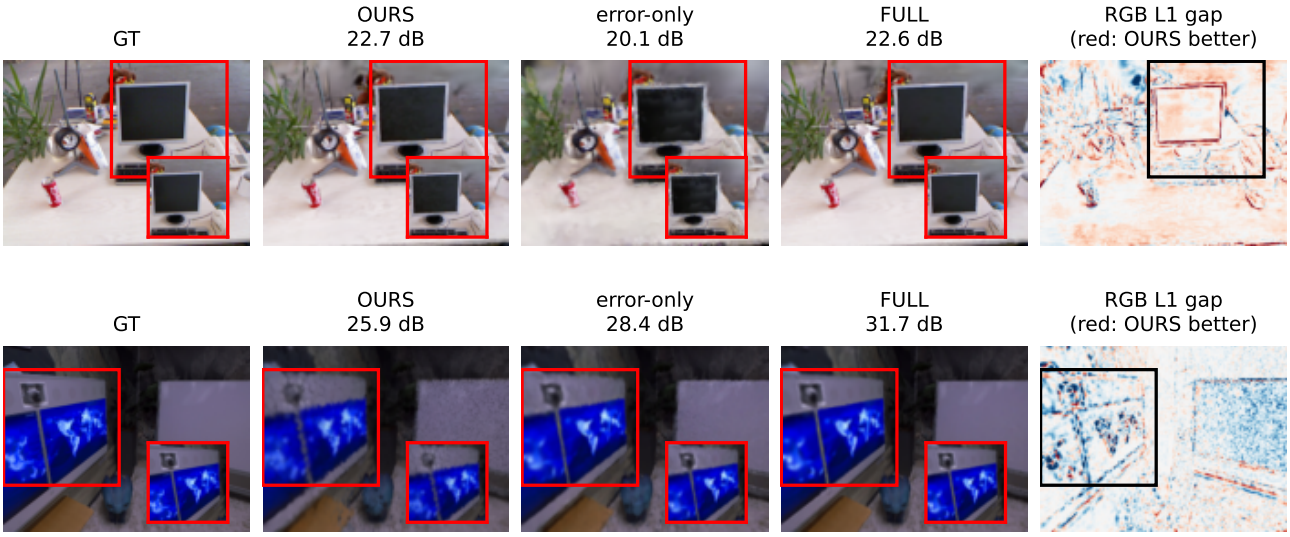


Figure 4. **Qualitative comparison across both evaluated substrates.** Top: an intermediate TUM `fr2_xyz` view (90), where OURS has lower per-view error than `error_only`; bottom: Replica `office0` view 98, where the ranking reverses. Columns show ground truth, OURS, `error_only`, FULL, and signed RGB L_1 error gap (red: OURS has lower error). Insets mark the same crop in each row. PSNR labels are per-view values from seed 42, shown only to illustrate the visual comparison; they are not sequence means.

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